



SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Assignment No. 05

Subject:	DevOps Lab
TCODE:	TE7950
Name of Student	Ria Vinod
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Branch	CSE, Batch (2023-27), L3
Academic Year & Semester	25 – 26, Sem VII
Date of Performance	16 th July, 2026
Title of Assignment:	Continuous Integration with Jenkins
Theory:	Theory Jenkins is an open-source automation server used to implement Continuous Integration (CI) and Continuous Delivery (CD). It automates software development tasks such as building, testing, and deploying applications. Jenkins integrates with version control systems like Git to automatically retrieve source code and execute predefined build steps. In this experiment, Jenkins was installed using Docker, configured with the Git plugin, and connected to a GitHub repository. A Freestyle project was created to pull source code from the repository and execute shell commands. The build was then triggered manually, and the console output was examined to verify successful execution. Commands Used • docker --version – Verified that Docker was installed on the system. • docker ps – Checked whether the Jenkins Docker container was running. • docker run -d -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home --name myjenkins jenkins/jenkins:lts-jdk17 – Created and started the Jenkins Docker container. • docker logs myjenkins – Displayed the Jenkins logs to obtain the initial administrator password during the initial setup.

- docker exec myjenkins cat /var/jenkins_home/secrets/initialAdminPassword – Retrieved the initial administrator password generated by Jenkins.
- Execute Shell Commands**
- echo "BUILDING: \${JOB_NAME}" – Displayed the Jenkins job name.
 - echo "BUILD NUMBER: \${BUILD_NUMBER}" – Displayed the current build number.
 - echo "WORKSPACE: \${WORKSPACE}" – Displayed the Jenkins workspace path.
 - ls -la – Listed all files present in the Jenkins workspace.
 - echo "Hello Jenkins!" – Displayed a sample message to verify successful execution of the build step.

Implementation:

Step 1: Verify Docker and Jenkins Container

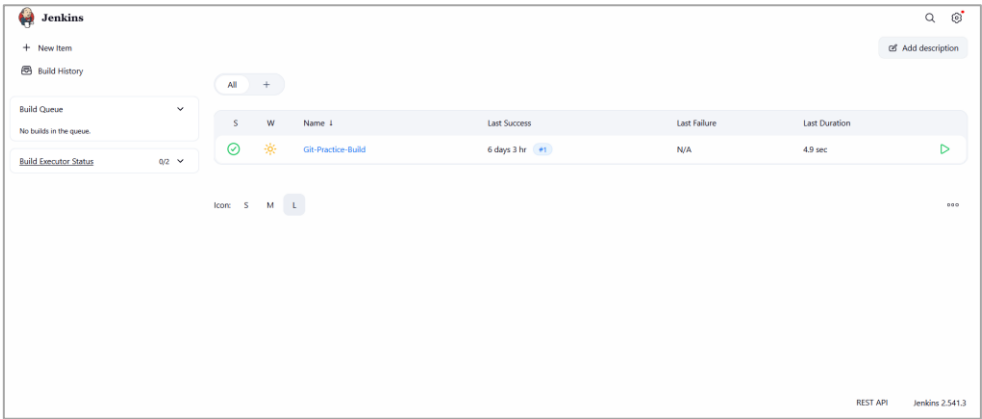
```
C:\Users\riavi>docker --version
Docker version 29.6.1, build 8900f1d

C:\Users\riavi>docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
defd1b9f4c1a   jenkins/jenkins:lts-jdk17          "/usr/bin/tini -- /u... 6 days ago    Up 44 seconds  0.0.0.0:8080->8080/
tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000/tcp, [::]:50000->50000/tcp   myjenkins

C:\Users\riavi>
```

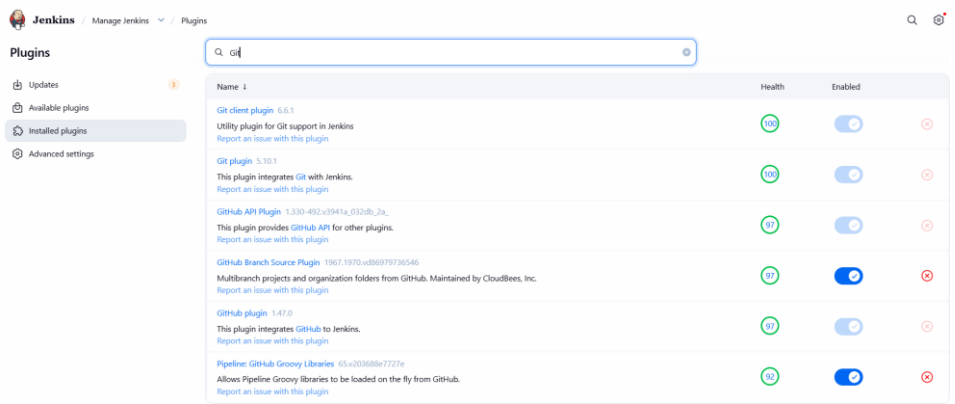
Verified Docker and Jenkins Container — Verified Docker installation and confirmed that the Jenkins Docker container (myjenkins) was running successfully.

Step 2: Access Jenkins Dashboard



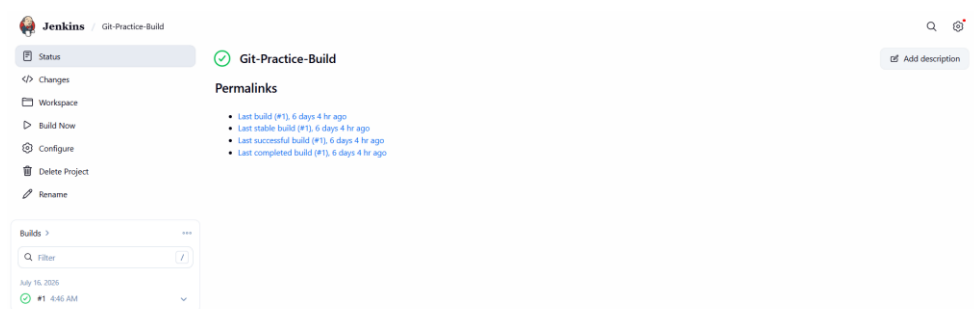
Accessed Jenkins Dashboard — Successfully accessed the Jenkins Dashboard and verified that the existing Freestyle project (Git-Practice-Build) was available.

Step 3: Verify Git Plugin Installation



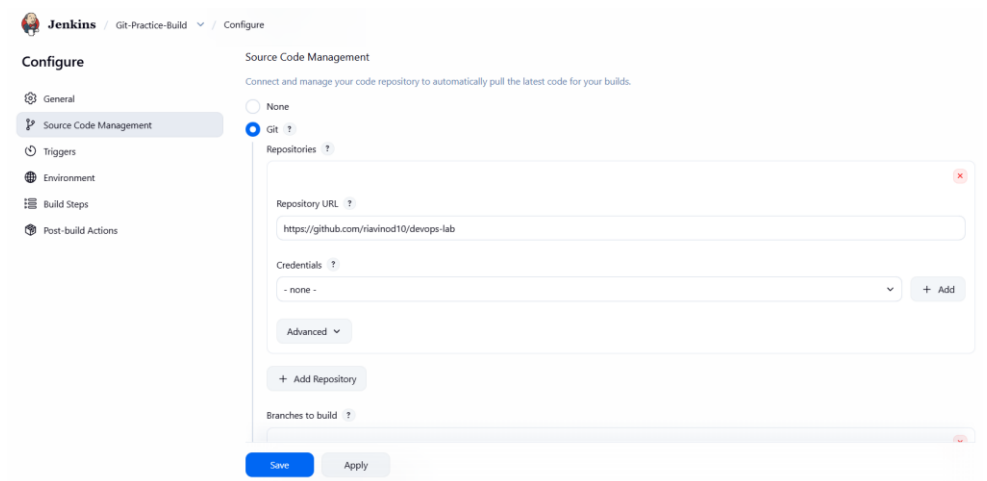
Verified Git Plugin — Verified that the Git plugin was installed and enabled, allowing Jenkins to integrate with Git repositories for source code management.

Step 4: Open the Freestyle Project



Opened Freestyle Project — Opened the existing Freestyle project (Git-Practice-Build) to verify its configuration and prepare it for executing the build process.

Step 5: Verify Source Code Management (Git)



Configured Source Code Management — Configured the Freestyle project to retrieve source code from the GitHub repository by selecting Git as the Source Code Management option and specifying the repository URL.

Step 6: Verify Build Step

Configure

- General
- Source Code Management
- Triggers
- Environment
- Build Steps**
- Post-build Actions

Build Steps

Automate your build process with ordered tasks like code compilation, testing, and deployment.

Execute shell ?

Command

See [the list of available environment variables](#)

```
echo "=====
echo "BUILDING: ${JOB_NAME}"
echo "BUILD NUMBER: ${BUILD_NUMBER}"
echo "WORKSPACE: ${WORKSPACE}"
echo "=====

echo "Repository Files:"
ls -la

echo "Hello Jenkins!"
```

Advanced ▾

+ Add build step

Save

Apply

Configured Build Step — Verified the Execute shell build step containing shell commands to display build information, list the repository files, and print a sample message during the build process.

Step 7: Save and Run the Build

Status

</> Changes

Workspace

Build Now

Configure

Delete Project

Rename

✓ Git-Practice-Build

Permalinks

- Last build (#1), 6 days 4 hr ago
- Last stable build (#1), 6 days 4 hr ago
- Last successful build (#1), 6 days 4 hr ago
- Last completed build (#1), 6 days 4 hr ago

Builds >

...

Filter

/

Today

✓ #2 9:09 AM

▾

July 16, 2026

✓ #1 4:46 AM

▾

Executed Build — Successfully executed the Freestyle project using Build Now and verified that the build completed successfully with a SUCCESS status.

Step 8: View Console Output

Console

```
Started by user unknown or anonymous
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/Git-Practice-Build
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/jenkins_home/workspace/Git-Practice-Build/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/riavinod10/devops-lab # timeout=10
Fetching upstream changes from https://github.com/riavinod10/devops-lab
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/riavinod10/devops-lab +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision 245b30d7eaabf375dd382f347da20725bfe4bda0 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 245b30d7eaabf375dd382f347da20725bfe4bda0 # timeout=10
Commit message: "Merge conflicting-feature, resolve README conflict"
> git rev-list --no-walk bb5e95a497d32e8989c1710866067967357019f6 # timeout=10
[Git-Practice-Build] $ /bin/sh -xe /tmp/jenkins16101747806234397119.sh
+ echo =====
+ echo BUILDING: Git-Practice-Build
BUILDING: Git-Practice-Build
+ echo BUILD NUMBER: 2
BUILD NUMBER: 2
+ echo WORKSPACE: /var/jenkins_home/workspace/Git-Practice-Build
WORKSPACE: /var/jenkins_home/workspace/Git-Practice-Build
+ echo =====
+ echo Repository Files:
Repository Files:
+ ls -la
total 16
drwxr-xr-x 3 jenkins jenkins 4096 Jul 22 09:09 .
drwxr-xr-x 3 jenkins jenkins 4096 Jul 16 04:46 ..
drwxr-xr-x 8 jenkins jenkins 4096 Jul 22 09:09 .git
-rw-r--r-- 1 jenkins jenkins 190 Jul 22 09:09 README.md
+ echo Hello Jenkins!
Hello Jenkins!
Finished: SUCCESS
```

Verified Console Output — Examined the console output and verified that Jenkins successfully cloned the Git repository, executed the configured shell commands, and completed the build with a SUCCESS status.

Subject Faculty Name: Prof. Arijit Dutta